Materials and Methods

Study design and data sources

We analyzed species with complete data for each analysis. The transposable-element (TE) panel contained 34 species and the microscopy and phylogenetic comparative panel contained 24, with 21 species shared between them. *Desmognathus brimleyorum*, *D. folkertsi*, and *D. ochrophaeus* had microscopy measurements but no TE data; their missing values were not imputed. Genomic and microscopy measurements were matched by species, without assuming that they came from the same animals.

Genome assemblies and raw sequencing data were obtained from the National Center for Biotechnology Information Assembly and Sequence Read Archive databases. We reconciled source names with the study taxonomy using identity records for each accession. The two available dnaPipeTE outputs for *D. orestes* were technical runs rather than biological replicates. We normalized each run separately and averaged their compositions with equal weight.

Alex Pyron provided maximum snout–vent length (SVL), body-size, and life-history data. We used maximum SVL rather than total length and retained the larger of the two reported values for *D. orestes* (52 and 50 mm). Reproductive strategy was classified as direct development, aquatic eggs, or terrestrial eggs with aquatic larvae. The TE panel contained three direct developers, seven aquatic-egg species, and 24 terrestrial-egg species with aquatic larvae. Because strategy assignments were supplied after the initial PCA had been inspected, these comparisons were exploratory. Three microscopy species lacked an egg-placement assignment in the fixed analysis dataset; their aquatic-larval development was known, and they remained in the comparative analyses.

Genomic read processing and de novo TE library construction

To remove mitochondrial reads, we indexed a mitochondrial reference (GenBank AY728227.1) and aligned paired reads with BWA-MEM using eight threads. After converting and sorting the alignments with SAMtools, samtools view -f 12 -F 256 retained pairs in which neither mate mapped to the mitochondrial reference. These pairs were the nuclear-read input for subsequent TE analyses.

A *Desmognathus*-specific repeat library was generated with RepeatModeler in the TEtools 1.87.sif container under Singularity 3.7.4. A database was built for each assembly, and RepeatModeler ran with 32 threads, -LTRStruct, and -genomeSampleSizeMax 10000000000. Redundant consensus sequences were clustered with CD-HIT-EST using -d 0 -aS 0.8 -c 0.8 -G 0 -g 1 -b 500 -T 64 -M 500000. The custom library was then used as the TE database for dnaPipeTE.

TE abundance estimation and classification

TE abundance was estimated with dnaPipeTE from mate 1 of the nuclear-filtered short reads. The analysis used Singularity 4.1.2, 16 CPUs, a configured 15-Gb genome-size denominator, 0.1× target coverage, RM_t = 0.15, two internal samples, and the custom TE library. The 15-Gb value controlled read sampling; it was not a measured genome size or an estimate of absolute TE load.

We assigned repeat components to class, order, and superfamily from the dnaPipeTE _reads_per_component_and_annotation output. At each taxonomic level, the aligned bases in each classified TE category were divided by the total classified aligned bases for that species. Total aligned bases, classified aligned bases, and unresolved aligned bases were recorded for quality control.

Unresolved mass did not enter the diversity calculations. The RepeatMasker .align files used for divergence profiles described dnaPipeTE Trinity repeat-contig assemblies rather than the accession-linked genome assemblies.

TE diversity and compositional ordination

TE diversity followed the top-ten scope of Wang et al. (2025). We selected the ten most abundant strictly positive TE superfamilies independently for each species, breaking ties by superfamily name. Their abundances were reclosed to sum to one. We then calculated natural-log Shannon entropy, $H' = -\Sigma\ p_i \ln(p_i)$, and the Gini-Simpson index, $1 - \Sigma\ p_i^2$, where p_i is the reclosed proportion of classified aligned bases assigned to superfamily i. Wang et al. (2025) do not state whether their retained top-ten abundances were reclosed; we specify that step here.

We used PCA to describe classified superfamily composition across the 34 TE species. After averaging technical runs, we retained superfamilies with positive abundance in every species, leaving 24 shared superfamilies. Ginger, Merlin, Dada, Chapaev, and CR1 were excluded because they were not detected in every species. Each composition was reclosed to sum to one and transformed by the centered log ratio, $CLR(p_i) = \ln(p_i) - \text{mean}[\ln(p)]$. No zero replacement was needed because the retained matrix contained no zeros. CLR columns were centered without variance scaling, and principal components were calculated by singular-value decomposition. For a reproducible axis orientation, the loading with the largest absolute value on each axis was set to be positive. This ordination provided supporting context for compositional variation and was separate from the eight main figures.

Each diversity index was regressed separately on conditional genome size for the 21 species shared by the TE and microscopy panels. Pagel-lambda phylogenetic generalized least squares (PGLS) used untransformed diversity and genome size in pg/1C, with lambda estimated by maximum likelihood between zero and one. Two-sided coefficient tests and 95% intervals used the *t* distribution with 19 residual degrees of freedom. The indices summarized the same selected composition and did not constitute independent replications. Genome-size measurement intervals were shown but were not propagated into the regressions.

Repeat divergence profiles

Repeat divergence profiles were reconstructed from RepeatMasker .align files for the dnaPipeTE Trinity repeat-contig assemblies. The inclusive length of each aligned hit was calculated as |query end - query start| + 1. We removed rows with invalid divergence or length values and grouped reported divergence into one-percentage-point bins with a terminal 50% bin. For each species, aligned bases in each TE category and divergence bin were divided by that species' total inclusive aligned RepeatMasker hit bases. Missing combinations of species, category, and bin were assigned zero before averaging across the 34 species with equal weight.

These fractions describe aligned repeat annotations; they do not measure genome occupancy, insertion counts, insertion ages, or evolutionary rates. Unclassified repeats were kept as an annotation category, not a biological TE order.

For each species, the low-divergence LTR score was the number of LTR aligned-hit bases in divergence bins 0–4 divided by all LTR aligned-hit bases across the available bins. The bin definition was fixed without a threshold search. We also calculated a fraction using all RepeatMasker aligned-hit bases as the denominator. This alternative equals the within-LTR score multiplied by the overall LTR fraction

of aligned-hit bases, allowing relative divergence-profile shape to be distinguished from total LTR representation. Both fractions were strictly between zero and one and were logit-transformed without a pseudocount.

In the 34 TE species, Pagel-lambda regressions related the within-LTR score to reproductive strategy, log10 maximum SVL, and both predictors together. An intercept-only model provided a reference. Lambda and residual variance were estimated separately by maximum likelihood for each equation. Omnibus strategy tests compared models differing by two group parameters with an approximate chi-square likelihood-ratio test using two degrees of freedom. Contrasts used aquatic eggs as the reference and were reported on the raw logit scale with *t* intervals based on residual degrees of freedom. Prediction curves extended only across each group's observed SVL range. Group sizes were not counts of independent evolutionary transitions; post-hoc strategy coding and phylogenetic clustering also limited interpretation.

Two predefined quality checks calculated unadjusted Spearman correlations between the within-LTR score and, separately, the dnaPipeTE unresolved fraction and the repeat-aligned fraction of its configured read sample. These descriptive comparisons assessed quality across pipelines and did not correct estimates of genome-wide TE abundance.

LTR terminal-to-internal depth proxy

DANTE, LTRharvest, and LTRdigest identified candidate full-length long terminal repeat (LTR) elements and annotated the 5' LTR, internal region, and 3' LTR. Candidate sequences were matched to TEsorter annotations by exact sequence identifier and coordinates. Eligible elements were at least 3,000 bp long, were classified by TEsorter as Order = LTR and Superfamily = Gypsy, and had at least five whitespace-delimited DOMAIN|MODEL annotations. These criteria identified 408 eligible LTR/Gypsy elements across 30 species.

Trimmed, mitochondrial-filtered paired reads from each species were mapped to its LTR contigs with Bowtie2 using -L 20 --very-sensitive-local. SAMtools converted, sorted, and indexed the alignments and calculated depth at each position. Positions absent from the depth output were assigned zero. Mean terminal depth was the length-weighted mean across the 5' and 3' LTRs; mean internal depth covered the annotated internal region. Dividing mean terminal depth by mean internal depth gave the terminal-to-internal ratio. Archived records do not preserve the Bowtie2 and SAMtools versions or the policies for mapping quality, multimappers, secondary or supplementary alignments, and duplicate reads.

Before summarizing species, we applied an inclusive two-sided Tukey filter to the untransformed element ratios within each species. Elements were retained when $Q1 - 1.5 \times IQR \leq ratio \leq Q3 + 1.5 \times IQR$. Quartiles and fences were calculated from the 408 eligible elements before coverage filtering; 381 elements passed this IQR rule. A separate coverage filter required positive depth at 80% or more of the expected positions in each of the 5' LTR, 3' LTR, and internal region. This filter retained 407 elements; its intersection with the IQR filter retained 380 elements across all 30 species. We summarized each species by its median ratio and calculated central 95% percentile intervals from 2,000 element-bootstrap replicates. Reproducible seeds were derived from the species, filter branch, and summary statistic using SHA-256.

We used terminal-to-internal depth as an indirect proxy for mapping patterns and possible deletion footprints, based on the hypothesis that recombination between terminal repeats can remove internal sequences. The internal region included the entire intervening sequence, including positions with zero

depth. A ratio of one indicates equal mapped terminal and internal depth; it is not a calibrated no-deletion baseline.

The genome-size comparison used one median ratio from each of the 19 species with both conditional genome size and primary LTR summaries, representing 239 retained elements. Pagel-lambda PGLS regressed log10 median ratio on log10 genome size, constraining lambda between zero and one. Two-sided tests and 95% coefficient intervals used 17 residual degrees of freedom. Element counts were neither regression weights nor independent biological replicates. Sensitivity analyses used Brownian covariance, ordinary regression, an untransformed PGLS, and 19 leave-one-species-out fits. Measurement error and tree uncertainty were not propagated. Converting between Gbp and pg/1C changes the intercept but leaves the log–log slope and inference unchanged.

Specimen preparation, staining, and image acquisition

Each microscopy specimen identifier denoted one animal. Multiple images and measured objects from that animal were repeated observations, not independent biological replicates. For specimens sampled directly, erythrocytes were collected with a pipette from an incision through the heart and deposited at the center of a microscope slide. Slides were stained with Feulgen-Schiff reagent following Hardie et al. (2002). All retained unknown slides and the two designated *D. fuscus* standard slides, from specimens 32469 and 32470, were stained together in one Feulgen run; the staining date remains unknown. The retained standard images were acquired at 40× by brightfield microscopy without immersion oil, using the green channel. The standard slides underwent the same fixation protocol as the other slides, although this does not establish identical prior preservation histories. All images used for morphometry and IOD were acquired by brightfield microscopy on an Olympus APX100 with a 40× objective. Subsequent images acquired with a 40× oil-immersion objective were excluded because their quality was poor.

Cell and nucleus segmentation and morphometry

Cellpose segmented cell boundaries, and a custom-trained YOLO model segmented nuclei. Masks retained identifiers for source images, tiles, centroids, overlaps, and individual objects. Eligible objects formed a valid one-to-one cell-nucleus pair, did not touch an image edge or overlap other cells or debris, and passed manual review.

Within each species, eligible cells were ranked from largest to smallest by cell area. We targeted the 50 largest accepted cells and replaced rejected objects with the next-largest eligible cell that passed review. This selection estimates the upper tail of detected eligible erythrocyte size rather than population-average cell size. The final morphometry dataset contained 1,152 reviewed cell-nucleus pairs from 24 species and 42 animals. Twenty-two species contributed 50 cells, *D. aeneus* contributed 44, and *D. wrighti* contributed 8.

Cell and nucleus areas were calculated from mask pixels with a linear calibration of 0.12 micrometers per pixel, equivalent to 0.0144 square micrometers per pixel. The median area of accepted objects summarized each species. Paired-object bootstrap resampling provided uncertainty intervals conditional on the observed images and animals; these intervals do not estimate population-level variation among animals.

Nuclear IOD measurement and conditional genome-size calibration

Pixel optical density was calculated as log10(background intensity / pixel intensity). Summing these values over the nuclear mask gave nuclear integrated optical density (IOD). Images were matched on log edge sharpness and log relative ring noise, excluding IOD, darkness, nucleus area, and cell area from the matching variables. We assessed balance using a maximum absolute standardized mean difference threshold of 0.10 and a maximum Kolmogorov-Smirnov distance threshold of 0.25. Species failing either threshold were flagged and retained.

The final IOD dataset contained 805 reviewed nuclei from 51 images, 50 animals, and 24 species. We calculated the median per-nucleus IOD in each image, then used the arithmetic mean of those image medians as the species estimate, giving images equal weight regardless of nucleus count. We estimated uncertainty with 2,000 hierarchical-bootstrap replicates, resampling images and then nuclei within images. Each species estimate and interval was divided by the median estimate across the 24-species panel to obtain a relative-IOD index.

Four species in the IOD dataset and seven in the morphology dataset had only one animal, precluding estimates of among-animal variability for those species.

Conditional genome size was calculated as target genome size (pg/1C) = (target equal-image IOD / *D. fuscus* equal-image IOD) × 16.462167689 pg/1C. The reference value came from the selected pooled *D. fuscus* assembly estimate, 16.1 Gbp / 0.978 Gbp per pg = 16.462167689 pg/1C (Myers et al. 2025). The denominator was the finalized *D. fuscus* species estimate from four reviewed target images, not the two designated standard images. Although the standards and unknowns shared one staining run, the standard nuclei had not completed the same eligibility, image-quality matching, and acceptance process as the target nuclei. The approximately 41% difference between the two standard-slide median IOD values also remained unexplained, so those measurements did not establish a validated calibration denominator. For each conditional genome-size interval, target bootstrap draws were divided by the corresponding *D. fuscus* draws and multiplied by the fixed anchor. These intervals included sampled IOD variation but excluded uncertainty in the assembly anchor and staining-run calibration. The resulting estimates were IOD-derived, assembly-anchored species summaries rather than independently validated C-values. IOD resampling used seed 20260710 with deterministic species offsets.

Phylogenetic comparative analyses

Alex Pyron supplied the focal time-calibrated *Desmognathus* phylogeny. We reconciled taxon names with the study species, retained exact unsuffixed representative tips, and pruned alternate tips and species without complete data using ape::drop.tip. Internal nodes with a single descendant after pruning were collapsed. Duplicate species tips were not permitted. The primary morphology and conditional-genome-size analyses used 24 tips. To assess differences between trees independently of taxon membership, we pruned the focal tree to the same 18-species set as a published 18-species topology and compared the two trees.

Positive cellular traits and maximum SVL were log10-transformed. For the body-size and TE-integration analyses, each cellular trait and SVL was standardized using the mean and sample standard deviation of the full 24-species panel. The same scaling was used for reduced cohorts and alternative animal summaries. In contrast, the original three-trait path sensitivities standardized traits within each fitted dataset. For cellular, body-size, and TE-integration equations, Pagel-lambda PGLS estimated lambda by maximum likelihood between 10^{-7} and one. Each species contributed one model observation.

The original pairwise analyses assessed associations with forward, reverse, and two-predictor equations. Coefficient tests used the *t* distribution with residual degrees of freedom, and the displayed cellular pairwise coefficient intervals used the 0.975 *t* quantile with 22 residual degrees of freedom. Supporting detailed three-trait regression and path-edge tables used normal-approximation intervals, estimate ± 1.96 standard errors. For the body-size, nuclear-allometry, TE-integration, and expanded body-role analyses, fitted bands and coefficient intervals used the 0.975 *t* quantile with n − p degrees of freedom, where p included the intercept. Fitted-mean bands used the full coefficient covariance matrix. Back-transformed log-scale fitted values represent conditional medians without an arithmetic-mean bias correction. The bands describe coefficient uncertainty rather than individual prediction intervals, condition on fitted lambda, and omit uncertainty in its estimation.

Nuclear–cell scaling and paired nuclear fractions

We examined nuclear–cell allometry by regressing log10 nucleus area on log10 cell area without standardizing the slope. The test of proportional scaling used (slope − 1) divided by its standard error and n − 2 residual degrees of freedom. Four fixed comparisons used either pooled species medians or arithmetic means of within-animal medians, in all 24 species or with *D. aeneus* and *D. wrighti* excluded. These conditional regressions did not correct predictor error or establish an errors-in-variables structural exponent.

Nuclear area fractions were calculated for individual cell–nucleus pairs. We took the median fraction within each animal and averaged those medians equally within species, rather than dividing aggregate medians. With seed 20260924, 1,000 hierarchical draws resampled animals within species and paired cells within each selected animal, preserving observed sample sizes. Each draw retained all 24 species and summarized each species by the arithmetic mean of sampled animal medians. All 1,000 allometry refits re-estimated lambda. Central 95% ranges for nuclear fractions and fitted slopes described conditional measurement sensitivity, separately from the model's *t* intervals. All 42 animals and 1,152 pairs were retained in the source cohort, but single-animal species could not estimate between-animal uncertainty.

Body size and animal-aware sensitivities

Maximum SVL represented the species maximum rather than the body length of a microscopy animal. We regressed genome size, cell area, and nucleus area separately on maximum SVL. Additional equations related cell area to genome size and SVL, nucleus area to cell area and SVL, and nucleus area to genome size, cell area, and SVL. We also fitted marginal cellular equations and two-predictor nuclear equations. Every equation was fitted to the 24-species panel and to a specified 22-species subset excluding *D. aeneus* and *D. wrighti*. These were the smallest sampled species and the only direct developers in the microscopy panel, so their exclusion changed body-size range, developmental composition, and taxon membership together. No species or threshold search was used. The comparison could neither separate these influences nor test a difference between slopes.

The animal-aware body-adjusted sensitivity summarized morphology by the median of within-animal medians. This estimand differed from the arithmetic mean of animal medians used for nuclear fractions and the allometry alternative. For IOD, we averaged image medians within each animal, then averaged animal estimates equally within species. Conditional genome size used the same fixed *D. fuscus* anchor, recomputing the denominator for each aggregation or draw.

Using seed 2026092409, we generated 250 measurement draws by resampling observed animals, images within each sampled animal, and rows within each sampled image. Sample sizes were preserved at each level, and cell and nucleus measurements remained paired. The assays were resampled independently even when animal identifiers overlapped, leaving cross-assay covariance unmodeled. All draws retained the 24 species and full-panel scaling. Another 24 leave-one-species-out fits used released summaries and pruned trees to assess species influence. These procedures evaluated the original four body-adjusted hypotheses; they did not provide resampling support for the later eight-class comparison. SVL and calibration uncertainty were not modeled.

Phylogenetic path hypotheses

The original three-trait path analysis compared conditional genome size, nucleus area, and cell area using 25 directed acyclic graphs in 11 Markov-equivalence classes. Ten classes had testable conditional-independence claims. The saturated class had none and was excluded from CICc ranking. Pagel-lambda models were primary, alpha was 0.05, and classes within two CICc units of the best class were considered competitive. A fully connected three-trait diagram illustrated the saturated hypothesis but was not scored or assigned a ranked class weight.

Sensitivities for the original three-trait comparison used 250 measurement-bootstrap datasets, 24 leave-one-species-out fits, and 200 published bootstrap trees on the 18-species overlap. Measurement draws resampled pooled paired cell–nucleus objects within species and, independently for IOD, images and then nuclei. They used seed 20260710 with deterministic species offsets and recomputed the *D. fuscus* calibration denominator in each draw. We compared evolutionary covariance under Brownian motion, Pagel lambda, and Ornstein–Uhlenbeck models with fixed and random roots. Simulation calibration used seed 20260710 to generate 800 datasets, with 100 replicates of four generating classes under independent and Brownian residual regimes. Multiplying IOD by a positive calibration constant has no effect after log transformation and standardization; this invariance does not validate the absolute genome-size scale.

A subsequent comparison placed SVL upstream of all three cellular traits in four hypotheses: a body-only fork, a genome–cell–nucleus bridge, a genome–nucleus–cell bridge, and a genome fork to cell and nucleus. Each hypothesis represented a graphical equivalence class. We also evaluated the original ten testable three-trait classes in the 24- and 22-species cohorts to distinguish cohort changes from body adjustment.

The expanded body-role comparison comprised eight distinct equivalence classes. Four used the upstream-SVL hypotheses. The other four allowed a genome–cell–nucleus connection with SVL linked downstream to cell area, a genome–nucleus–cell–SVL chain, SVL linked to genome size, or SVL disconnected from the cellular chain. We checked equivalence using each graph's undirected skeleton and unshielded colliders, enumerating orientations and rejecting cyclic graphs. The genome-first cell-branch representative had four equivalent acyclic orientations. Because these shared graphical conditional independences, they were not fitted as four additional hypotheses.

We fitted the eight classes separately to 24 and 22 species using fixed full-panel scaling. Conditional-independence tests primarily followed the order body size, genome size, cell area, nucleus area. A prespecified reverse order, nucleus area, cell area, genome size, body size, assessed sensitivity to regression direction for pairs without a directed path. Ancestral pairs kept the test direction implied by the graph. Fisher's C, parameter counts, CICc, and normalized weights were calculated from the tested

claims; weights could be compared only within the same cohort, test order, and candidate set. Structural equations were fitted directly with PGLS using common scaling. Reversing a phylogenetic regression could change its numerical results because residual covariance parameters were re-estimated. The expanded comparison included no saturated model, exhaustive four-node search, measurement bootstrap, or tree bootstrap. Neither graphical equivalence nor acceptable global fit identified causal direction.

TE–cellular integration

TE integration used a fixed set of regressions in the exact 21 species with complete TE, cellular, and maximum-SVL data. The primary within-LTR score and the alternative using all aligned hits as the denominator were logit-transformed and standardized across all 34 TE species. Cellular traits and SVL kept the full 24-species log10 scaling. For each TE score, genome size was regressed on the score alone and with SVL. Cell area was regressed on genome size and SVL, with and without the TE score; nucleus area was regressed on cell area, genome size, and SVL, with and without the TE score. The most complex nuclear diagnostic had four predictors and an intercept, leaving 16 residual degrees of freedom. Significance was not used to select a preferred equation or denominator.

We compared cellular coefficients from matched 21-species fits with the existing 24-species estimates and the TE-adjusted 21-species estimates to distinguish changes in species membership from covariate adjustment. Genome-size prediction curves held SVL at the 21-species geometric mean. These equations were exploratory conditional diagnostics and did not establish a TE-to-genome-to-cell causal network. Missing TE measurements were not imputed.

All displayed regression tests were unadjusted and exploratory. Pointwise model intervals excluded tree, trait-measurement, and calibration uncertainty except where a separately described measurement sensitivity addressed a specific component. Selection of large cells, incomplete animal replication, source-tree provenance, and unresolved staining calibration limited biological and causal interpretation.

Software and data availability

Downstream analyses used preserved RepeatModeler, dnaPipeTE, RepeatMasker, LTR mapping, and reviewed microscopy outputs. Processing included matching species identities, averaging technical runs, filtering, calculating diversity and CLR-PCA, fitting phylogenetic models, and generating figures. Phylogenetic analyses used R 4.4.3, ape 5.8.1, phylolm 2.6.5, and phylopath 1.3.1. Released LTR and reviewed-morphology tables include eligibility, exclusion, and quality-control fields for each row. Other raw and high-performance-computing outputs remain in frozen archival storage. Source paths, dimensions, column definitions, and SHA-256 checksums accompany the released files.